From Agriculture to Services: Development without Industrialization

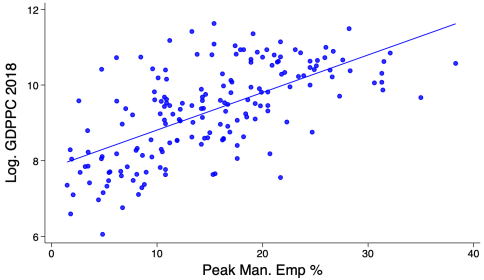
Tancredi Rapone[1,2]

UC3M October 2026

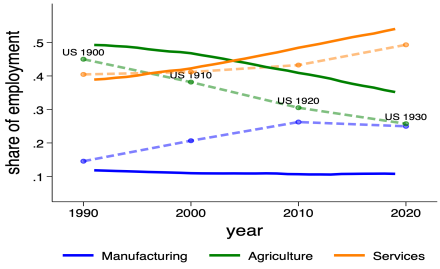
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Manufacturing Matters — But Many Countries Skip It



(a) Manufacturing peaks and income gaps



(b) Different paths into services.

Notes: In panel (b), dashed lines show the United States starting in 1900; solid lines show the unweighted average of low- and middle-income countries starting in 1990.

Sources: ILOSTAT, Gapminder.

VA

This Paper: Structural Change and Development

The traditional view of development

Kuznets (1966); Chenery and Syrquin (1975); Herrendorf, Rogerson and Valentinyi (2014)



A new development path

Rodrik (2016)



Does industrialization change the service economy that follows?

Mechanism: Manufacturing Shapes the Service Economy

Large-scale manufacturing

Coordination problems

hire →

High-skill producer services

Managers, engineers...
reusable methods.

diffuse →

Capabilities for services

Service firms adapt

Historical examples of reusable methods

DuPont

Return-on-investment accounting

General Motors

Multidivisional organization

Main idea: Manufacturing changes how production is organized.

Historical Example: Managerial Accounting

1914
DuPont
Manufacturing origin



ROI

1922–24
McKinsey
Codification



1935
Marshall Field's
Transfer to services



McKinsey becomes CEO

What I Do and What I Find

Document	**Cross-country stylized facts** Larger manufacturing peaks are associated with more high-skill producer services (HSPS) and faster growth after manufacturing declines.
Identify	**U.S. counties: shift-share evidence** Exposure to national growth in high-throughput manufacturing predicts larger industrial peaks and greater HSPS specialization in 2000.
Quantify	**Structural Change Model: Endogenous Skills and Growth** Scale creates demand for skills, skills lower cost of education and create growth. **98%** of the HSPS **40%** of the cross-country growth gradients.

Policy: Scale reforms strengthen the lasting effects of support for skills and organizational investment.

Literature

Literature and Contribution

- **Structural transformation and service-led development**

  Herrendorf et al. (2014); Rodrik (2016); Buera and Kaboski (2012); Buera et al. (2022); Fan et al. (2023); Huneeus and Rogerson (2024); Fujiwara and Matsuyama (2024); Sposi et al. (2026); Peters et al. (2026).

  ⇒ I study how manufacturing history shapes service composition and subsequent growth.

- **Firm scale, skills and organizational capital**

  Chandler (1977, 1990); Atkeson and Kehoe (2005); Cox (2025); Engbom et al. (2025); Gottlieb et al. (2026)

  ⇒ I link firm scale to skill demand and the creation of reusable organizational procedures.

- **Industrial policy: big push, increasing returns and persistence**

  Rosenstein-Rodan (1943); Hirschman (1958); Murphy et al. (1989); Juhász (2018); Incoronato and Lattanzio (2025); Choi and Levchenko (2025).

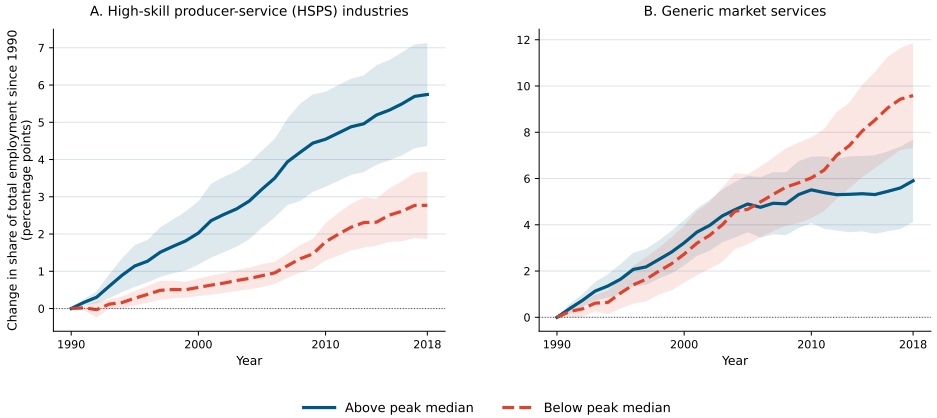
  ⇒ I show how industrial policy can have lasting effects **beyond manufacturing**.

Roadmap

1. **Cross-country stylized facts**
2. U.S. county evidence
3. Model: organization and growth
4. Quantitative analysis
5. Policy implications

Cross-Country Facts

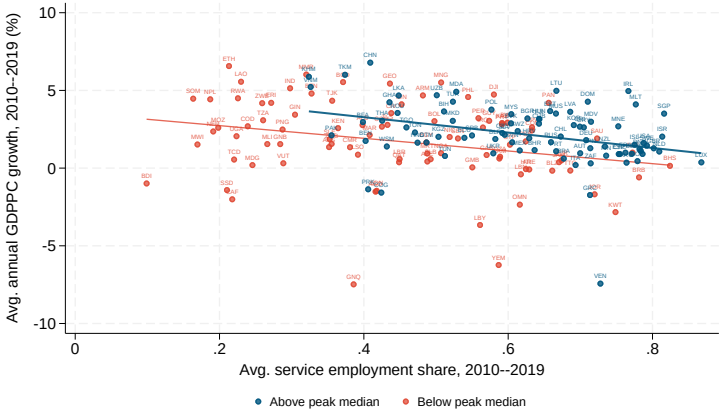
Stylized Fact 1: Manufacturing and Service Composition



51 economies; median manufacturing-peak split. Employment shares rebased to 1990; unweighted means ±1.96 SE. HSPS: business services, finance and real estate. Source: ETD and GGDC.

▸ Regression table ▸ Decomposition

Stylized Fact 2: Manufacturing and Service-Era Growth



2010–2019 averages; median manufacturing-peak split. Source: GGDC, ILOSTAT and World Bank/Gapminder.

▸ Regression table

U.S. County Evidence

Following Industrialization across U.S. Counties

Object	Measurement
Manufacturing peak	Peak manufacturing wage earners / classified employment, over 1880, 1900, 1920, 1930 and 1940.
HSPS in manufacturing	**Occupations**: accounting, coordination, personnel, engineering and technical work.
HSPS in services	**Industries**: information; professional, scientific and technical services; management of companies (NAICS 51, 54, 55).

Empirical test: Does HSPS appear first in manufacturing around the peak and later in Services as manufacturing declines?

▸ Definitions and sources

Tracing Capabilities around Manufacturing Peaks

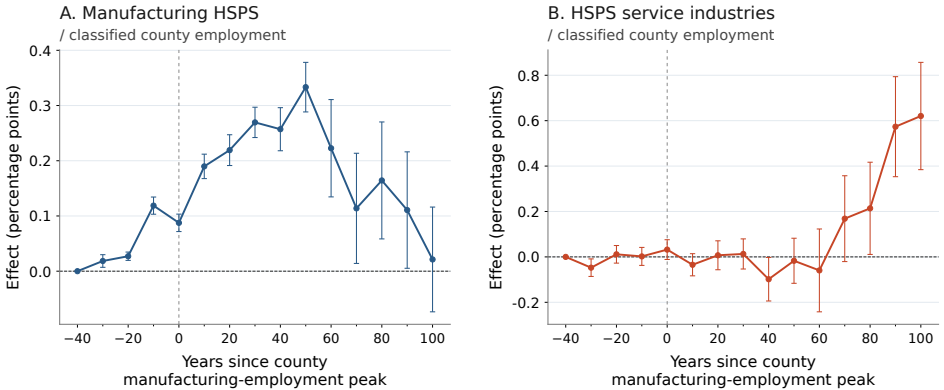
Y_{ct}: manufacturing HSPS or HSPS-service employment, both in % of classified employment.

$$Y_{ct} = \alpha_c + \gamma_{s(c),t} + \lambda_{t-P_c} + \sum_{k \neq -40} \beta_k \left[1\{t - P_c = k\} \frac{M_c^{\text{peak}}}{0.10} \right] + X_c' \delta \, \text{Trend}_t + \varepsilon_{ct}.$$

- **1,116 counties** peaking in 1900, 1920 or 1940, with a common forty-year pre-period.
- County, state-by-year and event-time fixed effects; baseline covariates interacted with linear calendar-time trends.

P_c: manufacturing-peak year. Reference: $t - P_c = -40$. Conley covariance, 100 km.

Manufacturing Capabilities Precede Specialized HSPS Services



Per 10 pp larger manufacturing peak; reference: −40. 1,116 counties; 95% CIs (Conley, 100 km). Source: Census and harmonized CBP.

▸ Falsification: other services ▸ By manufacturing type ▸ Channel accounting

Which Manufacturing Industries Require Production at Scale?

High-throughput manufacturing

Iron and steel; chemicals; machinery; processed foods.

Traditional low-throughput manufacturing

Textiles and apparel; wood and furniture; printing; leather goods.

Prediction: The service-sector legacy should be stronger when manufacturing expansion creates intensive organizational and technical demands.

Classification follows Chandler (1990).

▸ Pictures

National Industry Growth and Local Industrialization

Combine each county's **1880 industry mix** with subsequent **national industry growth**:

$$Z_c^g = \sum_{j \in g} \underbrace{\omega_{cj,1880}^g}_{\text{1880 industry share}} \underbrace{G_{j,-c,1880\text{–}1900}}_{\text{leave-county-out growth}}, \qquad g \in \{A, H, L\}.$$

- **Three instruments:** all manufacturing (A), high throughput (H), and traditional low throughput (L).
- **Treatment and outcome:** instrument the manufacturing peak to explain HSPS specialization within services in 2000.
- **Conditioning:** state and peak-cohort fixed effects, 1850 characteristics, and complete 1850–1870 development histories.

Identifying assumption: Conditional on having similar manufacturing and services sectors circa 1880 national industry growth affects services only through local industrialization.

National Industry Growth and Persistent HSPS Services

Dependent variable: HSPS share of private-service employment in 2000.

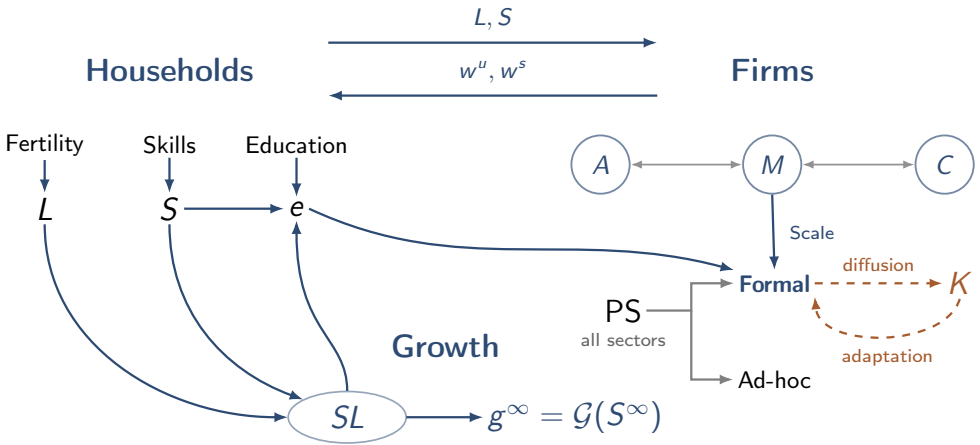
	OLS	All industries IV	High-throughput IV	Low-throughput IV
Manufacturing peak	0.290**	1.330	2.830**	−10.950
	(0.120)	(1.050)	(1.180)	(80.100)
First stage, per 1 SD of shifter		1.50***	1.93***	0.06
		(0.39)	(0.54)	(0.42)
First-stage Conley F		14.67	12.87	0.02
Kleibergen–Paap F		20.53	21.94	0.03
Anderson–Rubin p-value		0.203	0.009	0.596
Counties	1,487	1,487	1,487	1,487

State/peak-cohort FE, 1850 controls and 1850–1870 histories. Conley SE (100 km); AR: Anderson–Rubin. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: Census and QCEW.

▸ Rotemberg weights ▸ Measurement ▸ Shock inference ▸ Other checks

Model

Model Overview



Firm Scale and the Return to Formalization

$$\pi^*_{jt}(a,q) = \max_{\ell \geq 0} \left\{ p^v_{jt}(q) A_{jt} a \ell^\eta - \left[w^v_{jt} + c^o_{jt}(q) \right] \ell \right\}.$$

$p^v(q)$: value-added price net of intermediate costs; $c^o(q)$: coordination cost per unit of labor.

Proposition 1: Scale–formalization complementarity

$$\frac{\partial \pi^*_{jt}(a,q)}{\partial q} = n^*_{jt}(a,q) \Psi_{jt}(q)$$

$$\Psi_{jt}(q) = \underbrace{\frac{w^v_{jt} + c^o_{jt}(q)}{\eta \nu_{jt}} \frac{\partial \log p^v_{jt}(q)}{\partial q}}_{\text{productive-service cost effect}} + \underbrace{\frac{r p^L_t - p^H_t}{\bar{N}}}_{\text{coordination saving}}.$$

Conditional on equilibrium prices and $\Psi_{jt}(q) > 0$. If also $r p^L_t > p^H_t$, greater formalization raises desired employment.

▸ Derivation

Creating and Adapting Organizational Procedures

K_t: share of tasks with a publicly available procedure.

Create a new procedure

$$C_t^C(z) = \frac{p_t^H}{\bar{e}_t^o} e^{z-1}.$$

Adapt a public procedure

$$C_{jt}^A(z) = \frac{\rho}{\xi_j} C_t^C(z), \qquad \mathcal{F}(z) \leq K_t.$$

$\bar{e}_t^o$: education in organizational labor; ξ_j: adaptability; $0 < \rho < 1$, $\xi_m = 1$.

One-time investment; future operating returns

$$W_{jt}(a,s) = \pi_{jt}^*(a,s) + \max_{u \geq s} \left\{ -\mathcal{C}_{jt}^O(s,u;K_t) + \beta(1-\delta_f) W_{j,t+1}(a,u) \right\}.$$

Current production uses s; investment installs next period's frontier u.

▸ Investment and diffusion

Dynamics

Public organizational knowledge

$$K_{t+1} = \min\{1, K_t + \delta_K \mathcal{E}_t\}.$$

$\mathcal{E}_t$: employment-weighted exposure to privately created procedures not yet public.

Skill acquisition

$$p_t^S = \mathcal{S}(\Pi_{t+1}), \qquad \mathcal{S}' > 0.$$

p_t^S: skilled share of the entering cohort;
Π_{t+1}: anticipated skilled-wage premium.

Long-run growth

$$\bar{n}^\infty = 1, \qquad g^\infty = \mathcal{G}(S^\infty; \zeta), \qquad \frac{\partial \mathcal{G}}{\partial S} > 0.$$

Fertility settles at replacement; the terminal skilled share determines the sustained growth rate.

Terminal service economy; full-income ratio ζ held fixed. Conditions and derivation in the appendix.

▸ Diffusion and entry ▸ Transition equations ▸ Long-run growth

Calibration

Calibration: The U.S. Development Path

1850–2019; annual model. Joint SMM: 19 parameters, 100 moments in 13 blocks.

Parameter block	Principal empirical discipline
Demand and productivity	Sectoral employment, manufacturing peak, GDP and VA productivity growth
Fertility and skills	Fertility, college attainment, earnings premium and VA skill shares
Organization: R, ξ_a, ξ_c	Sectoral HSPS employment shares
Entry: F_a, F_m, F_c	Employees per employer firm in agriculture, manufacturing and services

Taking HSPS to the data: IPUMS + BEA Use Table

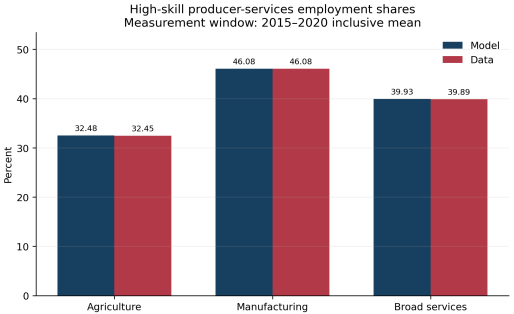
- Organizational and technical occupations; high skill: bachelor's degree or higher.
- Intermediate-service workers allocated to buyer sectors using BEA expenditure shares; final-consumer sales assigned to consumer services.

▸ Measurement ▸ Parameters ▸ Targeted fit ▸ U.S. paths

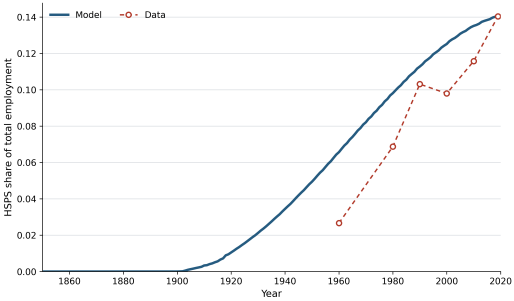
Fit and Validation

HSPS: Modern Fit and Historical Validation

Modern fit



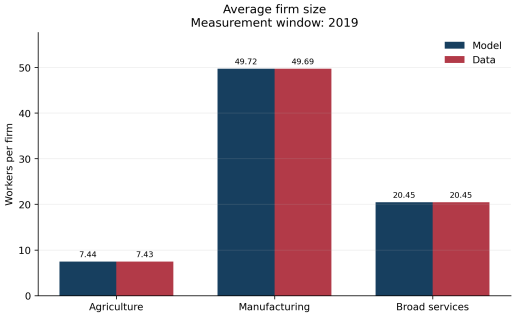
Historical validation



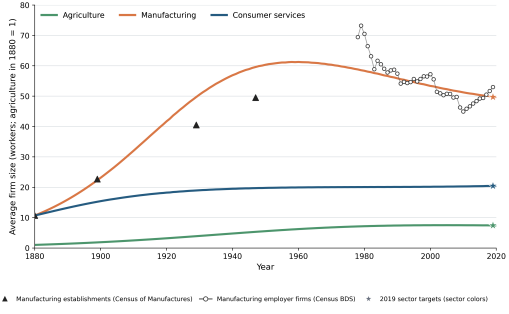
Left: targeted sectoral HSPS shares, 2015–2020 averages. Right: historical HSPS share of total employment. Source: IPUMS and BEA.

Firm Size: Modern Fit and Historical Validation

Modern fit

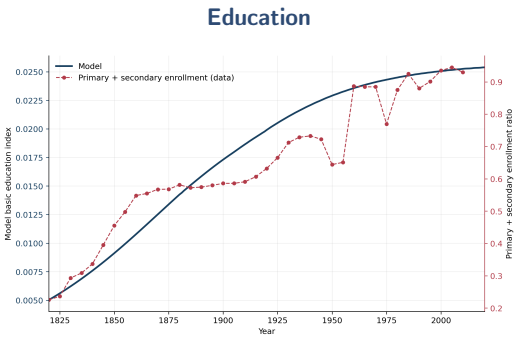


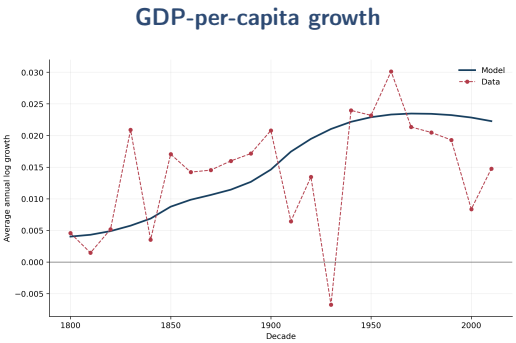
Historical validation



Left: targeted employees per employer firm, 2019. Right: historical manufacturing establishment and employer-firm sizes; initial scale anchored in 1880. Source: U.S. Census Bureau.

Untargeted Validation: Education and Growth





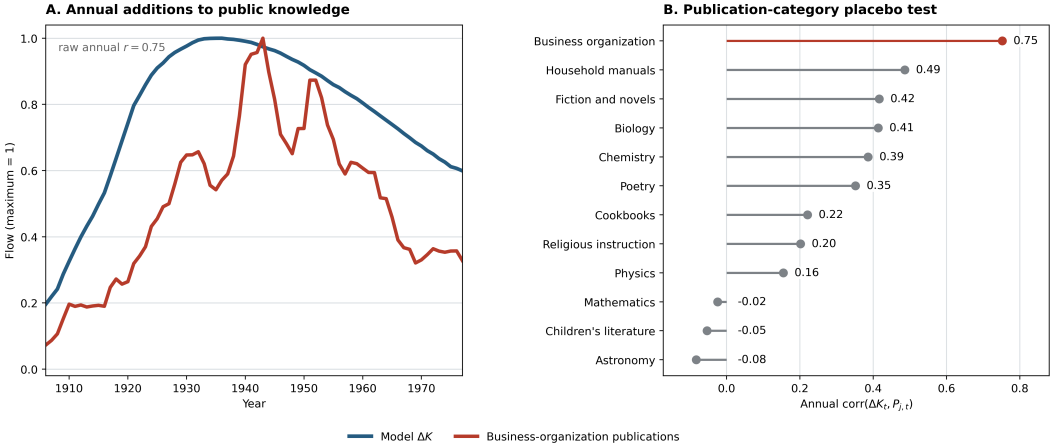
Left: model education index and primary–secondary enrollment. Right: annualized GDP-per-capita growth by decade, with the model shown as a smoothed trend. Neither series is directly targeted; GDP levels enter the calibration.

Measuring Public Organizational Knowledge

- Extract publications from the Library of Congress *Catalog of Copyright Entries*, 1906–1977.
- Count records containing fixed phrases/words per million OCR words. Compare to ΔK_t for various knowledge categories.

Publication category	**Examples of matched words and phrases**
Business organization	business systems; office management; personnel management; scientific management; time study; record-keeping...
Household manuals	household management; home economics; domestic science; housekeeping; household hints...
Chemistry	chemistry; chemical science; chemical analysis; biochemistry; electrochemistry...

Untargeted Validation: Public Organizational Knowledge



Copyright records per OCR word, 1906–1977; explicit manufacturing titles excluded. Lines: normalized five-year averages; correlations: annual data. Source: Library of Congress.

Quantitative Results

Counterfactuals: Manufacturing Histories and Development

1. **Vary manufacturing demand**
   Change only w_m throughout development; common initial conditions and all other parameters fixed.

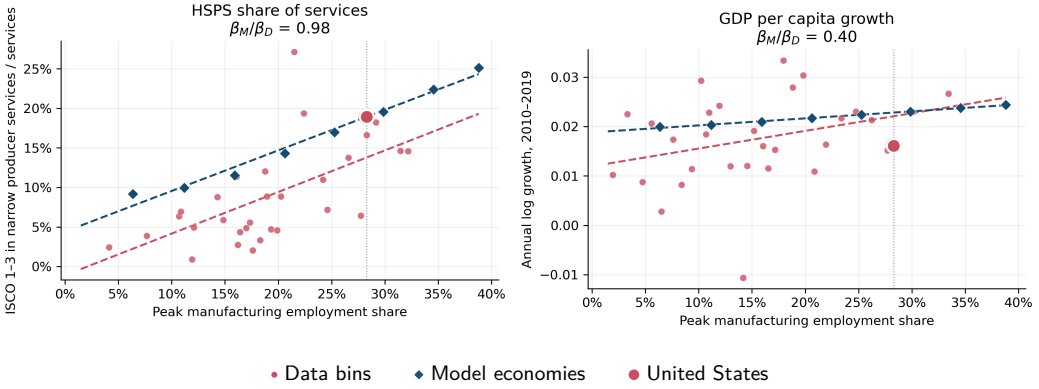
2. **Solve each development path**
   Manufacturing peaks, organizational knowledge, skills and income emerge in equilibrium.

3. **Compare model and data**
   Compare slopes relating each outcome to the historical manufacturing peak.

Outcomes: HSPS, tertiary attainment, income, GDP-per-capita growth and population growth.

▸ Demand-equivalent wedges

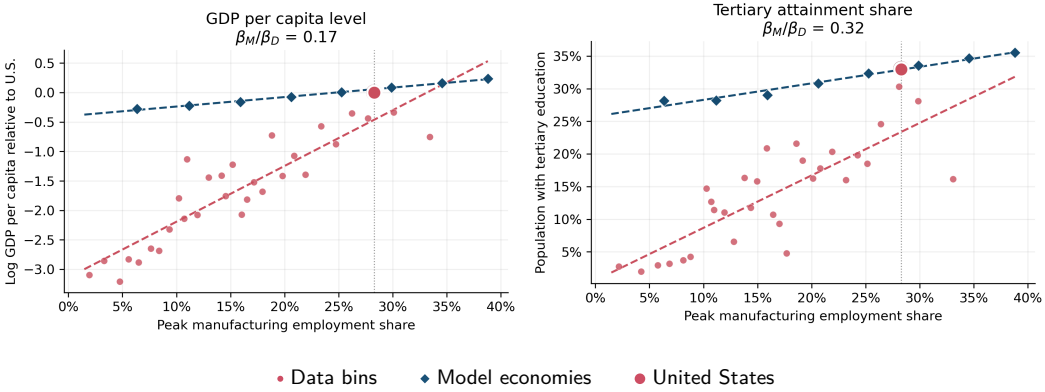
Manufacturing History, HSPS and Service-Era Growth



Data: 30 country bins; model: eight economies. Dashed lines: OLS on underlying observations; β_M/β_D: slope ratio. Samples: 48 countries (HSPS), 172 (GDP).

▸ HSPS measurement ▸ Sensitivity

Manufacturing History, Income and Education



Data: 30 country bins; model: eight economies. Income: 2019; tertiary attainment: 2010 and 2015 average. Samples: 172 countries (GDP), 137 (education). Sources: Gapminder and Barro–Lee.

Sensitivity | Mechanism accounting | Growth persistence

Industrial Policy

Temporary Support in Different Firm-Scale Environments

Start from the same inherited LMIC state in 2026.

Temporary interventions

Skill acquisition scholarships
Reduce the cost of becoming skilled.

Formalization support
Reimburse the cost of creating a new routine.

Mixed package
Split resources equally between the two.

Permanent firm-scale environments

1. Distortions remain
Keep $d_j = 0.41$.

2. Full reform
Reduce d_j to zero over 2026–2035.

3. Reform + U.S. entry costs
Also set F_m, F_c to their U.S. values.

Comparisons: manufacturing and HSPS wage subsidies.

2026–2045
Support expected to continue

→

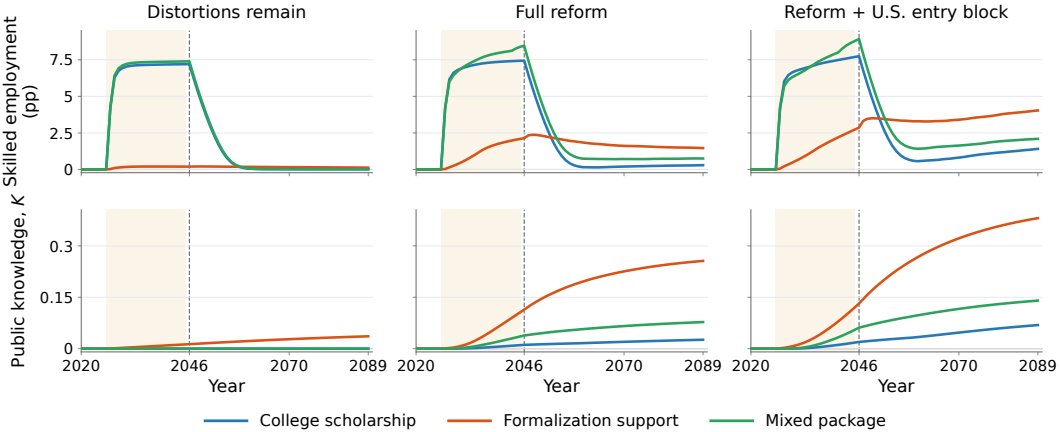
2046
Unexpected withdrawal

→

2089
Measure the legacy

▸ Model extension

What Remains after Policy Withdrawal?



Changes relative to each environment's no-support path. Shading: support in 2026–2045; dashed line: unexpected withdrawal in 2046.

▸ Formalization and income ▸ Policy outcomes ▸ U.S. entry block

Conclusion

Conclusion

1. **Industrial history shapes the service economy.**

   Larger manufacturing peaks predict more HSPS and faster subsequent GDP-per-capita growth.

2. **Production at scale builds transferable knowledge.**

   Firms develop reusable procedures that diffuse into services and sustain demand for skills.

3. **Scale reform and skills policies are complements.**

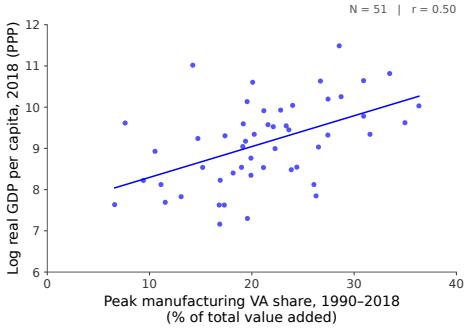
   Temporary support for skill acquisition and formalization has larger persistent effects when firms can grow.

Development without industrialization is possible,

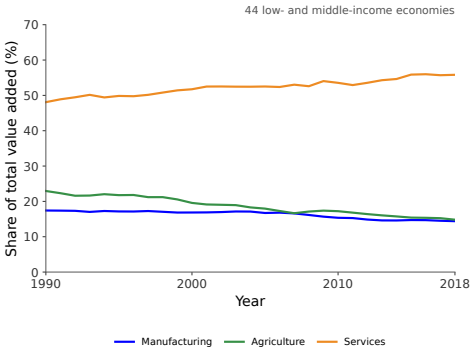
but the path into services matters.

Appendix

Motivation: Value Added



(a) Manufacturing VA peaks and income



(b) Sectoral value-added shares

Current-price VA shares, 1990–2018. Left: 51 economies; 2018 PPP income. Right: 44 low/middle-income economies, classified in 2018; unweighted means. Source: ETD and Gapminder.

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Manufacturing Peaks and Growth in Service Economies

Dependent variable: average annual log GDP-per-capita growth, 2010–2019 (pp).

	All countries		Services > 50%		Peak before 2000 and services > 50%
	(1)	(2)	(3)	(4)	(5)
Manufacturing peak (10 pp)	0.629**	0.398	0.645*	1.113**	0.942*
	(0.263)	(0.459)	(0.345)	(0.515)	(0.490)
Mean service share (10 pp)	−0.572***	−0.540**	−1.045**	−1.313**	−1.450**
	(0.206)	(0.215)	(0.438)	(0.605)	(0.555)
Mean manufacturing share (10 pp)		0.354		−0.924	−1.199
		(0.536)		(0.883)	(0.730)
Region and income-group FE	Yes	Yes	Yes	Yes	Yes
Institutional quality control	Yes	Yes	Yes	Yes	Yes
Observations	173	173	104	104	90
R^2	0.360	0.362	0.278	0.288	0.372

HC3-robust standard errors in parentheses. Employment shares are averaged over 2010–2019. Column (5) additionally restricts recorded peaks to years before 2000. Income groups: 2009 GNI. Institutions: control-of-corruption percentile. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Source: GGDC, ILOSTAT and World Bank.

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Manufacturing Peaks, Human Capital, and Service Composition

Dependent variable: average annual log GDP-per-capita growth, 2010–2019 (pp).

	Baseline (1)	HCQ (2)	HSPS (3)	Both (4)
Manufacturing peak (10 pp)	0.582*	0.264	0.340	0.167
	(0.343)	(0.315)	(0.303)	(0.305)
HCQ, 2015		0.841**		0.658**
		(0.351)		(0.325)
HSPS, 2018 (10 pp)			0.907*	0.624
			(0.493)	(0.455)
Mean service-employment share	Yes	Yes	Yes	Yes
Region and income-group FE	Yes	Yes	Yes	Yes
Observations	79	79	79	79
R^2	0.304	0.365	0.353	0.385
Peak-coefficient attenuation	—	54.7%	41.5%	71.4%

HC3 standard errors in parentheses; common sample of 79 countries. Service share: 2010–2019 average. Income groups: 2009 GNI. HSPS: ILO finance, business services and real estate / services excluding public administration, education and health. Attenuation relative to (1); joint HCQ–composition test in (4): $p = 0.028$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

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County HSPS Measures and Employment Denominators

Inside manufacturing: occupations

Accounting and record keeping; purchasing, logistics and coordination; personnel and labor relations; technical and scientific work.

In services: industries

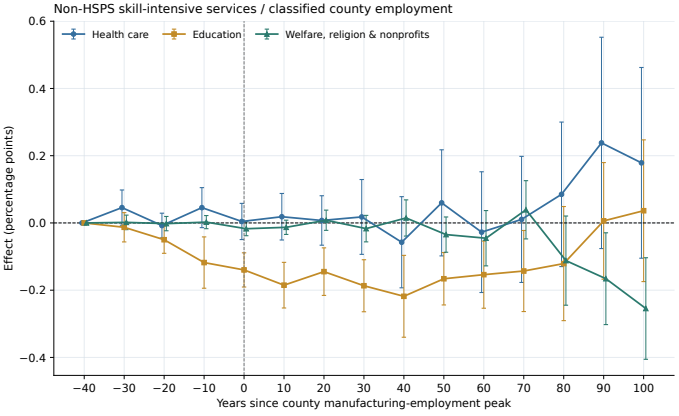
Information (NAICS 51); professional, scientific and technical services (54); management of companies and enterprises (55). Historical Census and SIC data use functional concordances.

Exercise	Denominator
Main event study	Classified employment: primary, manufacturing and services, reconstructed from the numerator's source in each vintage.
2000 cross section	QCEW private-service employment, excluding utilities.

Sources: Census of Manufactures, Population Census, CBP/EFSY and QCEW. Fixed 1850 county geography.

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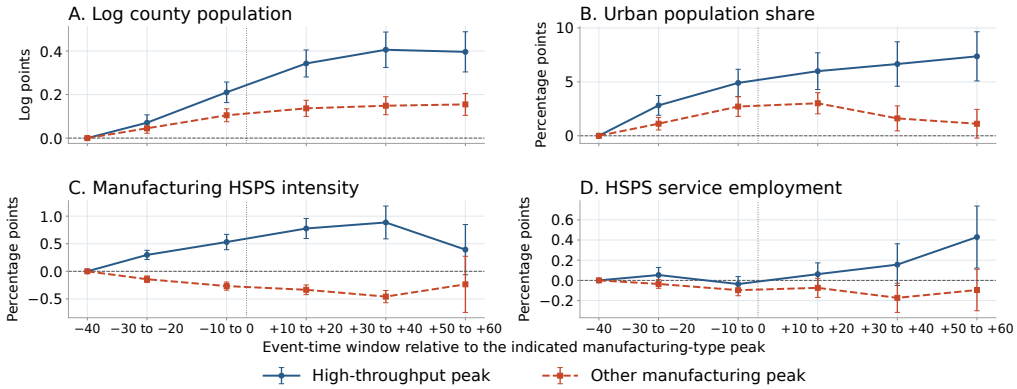
Falsification: Other HS Services



Service employment HS industries / classified employment. Same sample and specification as the HSPS-service panel; 95% CIs (Conley, 100 km). Source: Census and CBP.

◂ HSPS event study

Manufacturing Type: Development and HSPS Employment



Per 10 pp larger type-specific peak; 95% CIs (Conley, 100 km). HSPS denominators: manufacturing employment (C); classified private employment (D). Source: Census and CBP.

◂ Main event study ◂ Manufacturing types

Disclosure Treatment and Labor-Market Geography

High-throughput IV estimates under alternative outcome constructions.

Outcome and geography	N	Estimate	SE	First-stage F	AR p
QCEW public, counties, 2000	1,487	0.283**	0.118	12.87	0.009
QCEW flag-aware, counties, 2001	1,511	0.187*	0.100	11.25	0.066
EFSY native, counties, 2000	1,487	0.103	0.080	12.87	0.202
QCEW flag-aware, CZs, 2001	391	0.324**	0.141	9.03	0.005
EFSY harmonized, CZs, 2001	391	0.204	0.125	9.03	0.094
EFSY harmonized, CZs, 2010	391	0.279**	0.135	9.03	0.009

Preferred controls; Conley SE (100 km). EFSY: imputed CBP employment. CZ: commuting zone. AR: Anderson–Rubin. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Back

Scale, Manufacturing HSPS, and Later Service Composition

Dependent variable: HSPS share of private-service employment in 2000.

	Baseline	+ Scale	+ Mfg. HSPS	+ Both
Manufacturing peak	0.480***	0.240****	0.300***	0.150
	(0.100)	(0.120)	(0.110)	(0.120)
Log workers per manufacturing establishment		0.600***		0.410*
		(0.220)		(0.230)
Manufacturing HSPS / classified employment			2.112***	1.927***
			(0.529)	(0.535)
Attenuation of peak coefficient (%)	0.0	50.3	37.4	68.3

1,487 counties; state/peak-cohort FE and 1850 controls. Conley SE (100 km). $^{*}p < 0.10$, $^{****}p < 0.05$, $^{***}p < 0.01$. Source: Census and QCEW.

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Inference with Common Industry Shocks

Outcome and inference	Estimate	SE	p-value	First-stage F	AR p
High-throughput instrument					
Public QCEW, 2000: county Conley	0.283	0.118	0.016	12.87	0.009
Public QCEW, 2000: exact leave-out	0.283	0.159	0.088	5.26	0.062
Public QCEW, 2000: common shocks	0.284	0.132	0.041	3.94	0.047
Flag-aware QCEW, 2001: exact leave-out	0.187	0.121	0.135	5.02	0.065
Flag-aware QCEW, 2001–03: exact leave-out	0.164	0.108	0.142	5.39	0.082
All-industry instrument					
Public QCEW, 2000: exact leave-out	0.133	0.223	0.551	1.59	0.623

Exact leave-out: industry-clustered scores. Common shocks: leave-sample-out growth. 26 high-throughput industries; effective count: 4.71. Tests use finite-industry degrees of freedom. AR: Anderson–Rubin.

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Classification, Exposure, and Conditioning Checks

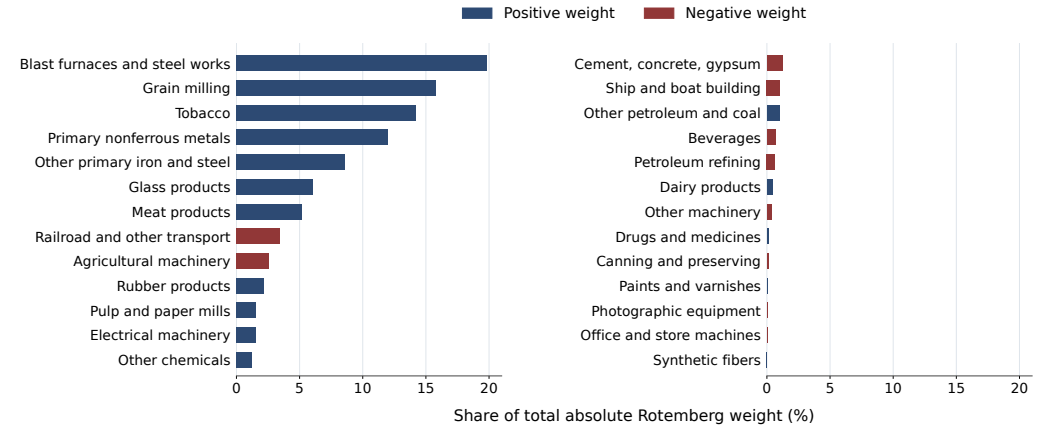
High-throughput IV; QCEW public-record outcome in 2000.

Specification	Estimate	SE	p-value	First-stage F	AR p
Exclude processed foods	0.325	0.257	0.207	3.63	0.159
Exclude machinery	0.189	0.098	0.055	28.98	0.041
Narrow continuous/process set	0.183	0.078	0.018	29.69	0.011
Add historical geography controls	0.320	0.120	0.008	13.43	0.002
Add manufacturing employment share in 2000	0.196	0.096	0.042	14.81	0.042
Preferred controls; Conley radius 500 km	0.283	0.170	0.096	2.83	0.067

Preferred controls except where indicated. Conley SE, 100 km unless stated otherwise. AR: Anderson–Rubin.

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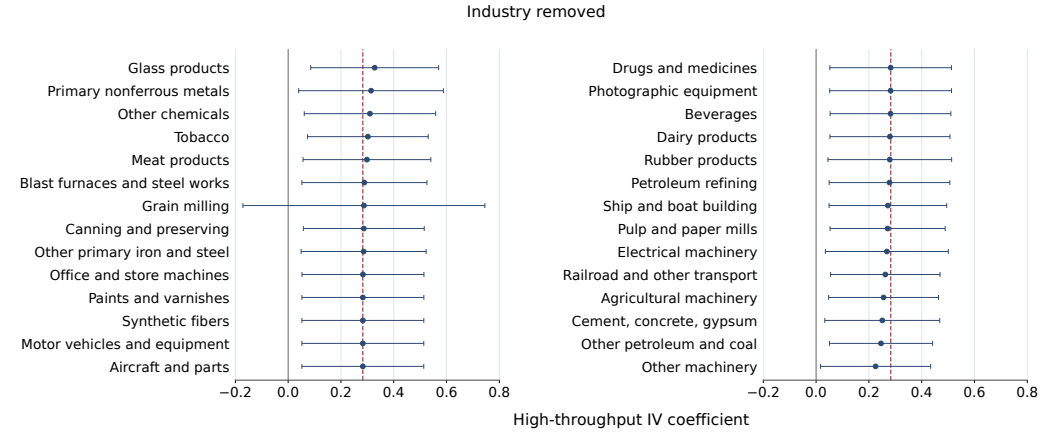
Which Industries Identify the High-Throughput IV?



26 industries; absolute Rotemberg weights after residualizing on the preferred controls. The five largest account for 70.4%; negative weights account for 10.2% of total absolute weight. Source: Census and QCEW.

Leave one out Back

High-Throughput IV: Leave One Industry Out



Each row drops the named industry and rebuilds the shifter. 2SLS estimates with 95% Conley intervals (100 km); preferred controls. The dashed line is the benchmark. Source: Census and QCEW.

◂ Weights ◂ IV results

Manufacturing Peaks and HSPS Employment Growth

Dependent variable: change in HSPS share of total employment, 1990–2018 (pp).

	(1)	(2)	(3)	(4)
Manufacturing peak (10 pp)	2.867***	2.937***	2.173***	2.283**
	(0.507)	(0.501)	(0.782)	(0.949)
GDP-per-capita gain ($\Delta \ln y$)		−1.202	−0.176	−0.145
		(0.732)	(1.029)	(1.079)
Service share, 1990 (pp)				−0.021
				(0.068)
Region and initial income-group FE	No	No	Yes	Yes
Economies	51	51	51	51
R^2	0.530	0.550	0.617	0.618

HSPS: business services, finance and real estate. Unweighted OLS; HC3 standard errors in parentheses. Income groups use 1990 GNI. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Sources: ETD, GGDC and Maddison (2023).

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Production, Producer Services and the Scale Result

$$\ell = \left[\alpha_j (h_s L^s)^{\frac{\sigma_L - 1}{\sigma_L}} + (1-\alpha_j)(h_u L^u)^{\frac{\sigma_L - 1}{\sigma_L}}\right]^{\frac{\sigma_L}{\sigma_L - 1}}.$$

$$P^p = \left[q^{1/\sigma_P}(H^p)^{\frac{\sigma_P - 1}{\sigma_P}} + (1-q)^{1/\sigma_P}(L^p)^{\frac{\sigma_P - 1}{\sigma_P}}\right]^{\frac{\sigma_P}{\sigma_P - 1}}, \quad \omega^P(q) = \left[q(p^H)^{1-\sigma_P} + (1-q)(p^L)^{1-\sigma_P}\right]^{\frac{1}{1-\sigma_P}}.$$

$$\ell^*(a,q) = \left[\frac{\eta p^v(q) A a}{w^v + c^o(q)}\right]^{\frac{1}{1-\eta}}, \qquad n^*(a,q) = \nu \ell^*(a,q).$$

The envelope theorem and the labor first-order condition imply

$$\frac{\partial \pi^*}{\partial q} = v^* \frac{\partial p^v}{\partial q} + \frac{n^*}{\bar{N}}(r p^L - p^H) = n^* \left[\frac{w^v + c^o(q)}{\eta \nu} \frac{\partial \log p^v(q)}{\partial q} + \frac{r p^L - p^H}{\bar{N}}\right].$$

Sector and time subscripts suppressed. Competitive producer-service supply gives $p^H = w^s / A^H$ and $p^L = w^u / A^L$.

◂ Scale result

Organizational Investment, Diffusion and Entry

$$\mathcal{C}^{O}_{jt}(s,u;K_t)=\int_s^u \widetilde{C}_{jt}(z;K_t)\,d\mathcal{F}(z),\qquad \widetilde{C}_{jt}(z;K_t)=\begin{cases}\min\{C^{C}_{t}(z),C^{A}_{jt}(z)\}, & \mathcal{F}(z)\le K_t,\\ C^{C}_{t}(z), & \mathcal{F}(z)>K_t.\end{cases}$$

$$\mathcal{E}_t=\frac{1}{N_t}\sum_j\int n^{*}_{jt}(a,s)\left[\int_{\mathcal{F}(z)>K_t} c_{jt}(a,s;z)\,d\mathcal{F}(z)\right]d\mu_{jt}(a,s).$$

$c_{jt}(a,s;z)$ indicates a privately created procedure still held by the firm; exposure counts only procedures not yet public.

$$N_t=\sum_j\int n^{*}_{jt}(a,s)\,d\mu_{jt}(a,s),\qquad K_{t+1}=\min\{1,K_t+\delta_K\mathcal{E}_t\}.$$

$$I_{jt}\ge 0,\quad F_j I_{jt}^{\phi_E}-V^{E}_{jt}\ge 0,\quad I_{jt}\left[F_j I_{jt}^{\phi_E}-V^{E}_{jt}\right]=0.$$

$$V^{E}_{jt}=\int W_{jt}(a,0)\,dG(a),\qquad M_{jt}=(1-\delta_f)M_{j,t-1}+I_{jt},\qquad \overline{n}_{jt}=\frac{N_{jt}}{M_{jt}}.$$

◂ Investment ◂ Dynamics

Aggregate Dynamics: Annual Implementation

Cohort replacement

$$p_t^S = \mathcal{S}(\Pi_{t+1}), \quad \mathcal{S}' > 0, \qquad S_{t+1} = \left(1 - \frac{1}{\Delta}\right) S_t + \frac{1}{\Delta} p_t^S.$$

Population

$$L_{t+1} = L_t \bar{n}_t^{1/\Delta}, \qquad \bar{n}_t = \sum_i \lambda_{i,t} n_{i,t}.$$

Sectoral and aggregate productivity growth

$$g_{j,t}^A = \left[1 + \lambda_A (S_t L_t)^{\phi_j}\right]^{1/\Delta} - 1, \qquad g_t = \sum_{i \in \{a,m,s\}} \lambda_{i,t} g_{i,t}^A.$$

Δ: generation length; $\lambda_{i,t}$: employment shares; Π_{t+1}: anticipated skilled-wage premium. Consumer and producer services share ϕ_c; $0 < \phi_j < 1$.

Households and skill choice | Equilibrium | Dynamics

Household Decision Rules and Skill Acquisition

Fertility and continuous education

$$e^*_{i,t} = \left[\sqrt{g_t\left(\frac{\tau_i}{\tau_e(S_t)} - \tau_h\right)} - \tau_h\right]_+, \qquad n_{i,t} = \frac{\gamma\zeta_t}{\tau_i + \tau_e(S_t)e^*_{i,t}}, \qquad \zeta_t = 1 + \frac{d_t}{\bar{w}_t}.$$

$$\bar{n}_t = \sum_i \lambda_{i,t} n_{i,t}, \qquad \tau_e(S_t) = 1 - S_t(1 - \tau^s_e).$$

Discrete skill choice of entering cohorts

$$p^S_t = \left[1 + \exp\left(-\frac{\beta_\pi \log \Pi_{t+1} - c^{\text{skill}}}{\sigma_{\text{skill}}}\right)\right]^{-1}.$$

c^{skill}: skill-acquisition cost; σ_{skill}: dispersion of idiosyncratic choice costs.

$[x]_+ = \max\{x, 0\}$. Education rule shown under $\tau_i/\tau_e(S_t) > \tau_h$; firms' organizational education input uses education chosen 30 years earlier.

◂ Model overview ◂ Dynamics

Perfect-Foresight Competitive Equilibrium

Given initial population, skills, knowledge, productivities and firm distributions:

1. Households choose consumption, fertility, education and cohort skill.
2. Firms choose inputs, employment and next-period formalization.
3. Entry satisfies free entry; producer-service suppliers are competitive.
4. Final goods, both producer services and both labor markets clear.
5. Aggregate states and firm distributions follow their laws of motion.
6. Anticipated aggregate prices and states equal their realizations.

$$\mu_{j,t+1}(B) = (1-\delta_f)\int 1\{(a, u^*_{jt}(a,s)) \in B\}\, d\mu_{jt}(a,s) + I_{j,t+1}\int 1\{(a,0)\in B\}\, dG(a).$$

Manufacturing is the numeraire. Idiosyncratic ability and exit risk; no aggregate uncertainty. Transition solved by backward firm optimization and forward aggregation.

◂ Dynamics

Population and Long-Run Growth

For a terminal skilled share S, fertility converges to replacement: $\bar{n}^{\infty} = 1$.

$$g^{\infty} = \mathcal{G}(S;\zeta) = \underbrace{\frac{(\gamma\zeta - \tau_s)^2}{\tau_s}}_{\text{Demographics}} \times \underbrace{\frac{1}{\tau_e(S)}}_{\text{Education efficiency}}.$$

Population adjusts to a unique finite level:

$$L^{\infty}(S;\zeta) = \frac{1}{S}\left[\frac{[1+\mathcal{G}(S;\zeta)]^{\Delta} - 1}{\lambda_A}\right]^{1/\phi_c}.$$

$$\tau_e(S) = 1 - S(1-\tau_e^s), \qquad 0 < \tau_e^s < 1 \quad \Longrightarrow \quad \frac{\partial \mathcal{G}(S;\zeta)}{\partial S} > 0.$$

Terminal service economy; $\tau_h = 0$, $\gamma\zeta > \tau_s$, and fixed full-income ratio $\zeta = 1 + d^{\infty}/\bar{w}^{\infty}$. Δ: generation length.

▸ Proof ▸ Manufacturing legacy ◂ Dynamics

Proposition 2: Replacement Fertility and Growth

Fix S and ζ in the terminal service economy; set $\tau_h = 0$.

$$g(L) = \left[1 + \lambda_A (SL)^{\phi_c}\right]^{1/\Delta} - 1, \quad e(L) = \sqrt{\frac{\tau_s g(L)}{\tau_e(S)}}, \quad n(L) = \frac{\gamma\zeta}{\tau_s + \tau_e(S) e(L)}.$$

$g(L)$ is increasing and $n(L)$ decreasing, with $n(0) > 1$ and $n(\infty) = 0$.

$$T(L) = L n(L)^{1/\Delta}, \qquad 0 < 1 - \frac{\phi_c}{2\Delta} < \frac{d \log T(L)}{d \log L} < 1.$$

At the unique fixed point, $n = 1$:

$$e^{\infty} = \frac{\gamma\zeta - \tau_s}{\tau_e(S)}, \qquad g^{\infty} = \frac{(\gamma\zeta - \tau_s)^2}{\tau_s \tau_e(S)}, \qquad \frac{\partial g^{\infty}}{\partial S} > 0.$$

$\gamma\zeta > \tau_s$, $0 < \phi_c < 1$ and $\Delta \geq 1$. The population map is increasing and moves toward the fixed point from either side.

◂ Proposition 2

Manufacturing and Long-Run Growth

Economy A Agriculture $\longrightarrow$ Manufacturing $\longrightarrow$ Services

Economy B Agriculture $\longrightarrow$ Services

$$g_A^{\infty} > g_B^{\infty} \iff S_A^{\infty} > S_B^{\infty} \iff K_A^{\infty} > K_B^{\infty}$$

$$\iff \underbrace{\frac{\tau_m}{\tau_s}}_{\text{education}} \underbrace{(S_1 L_1)^{\phi_m - \phi_c}}_{\text{productivity growth}} \underbrace{\left(\frac{F_m}{F_c}\right)^{\frac{2}{1+\phi_E}}}_{\text{firm scale}} > 1.$$

Common initial state; three-stage comparison with symmetric firms, common prices and terminal full-income ratio, and nonsaturated service formalization; $\tau_h = 0$, $\Delta = 1$.

▸ Proof and conditions ◂ Long-run growth

Proposition 3: The Manufacturing Legacy

Under the three-stage comparison:

$$K_{A,3} > K_{B,3} \quad \Longleftrightarrow \quad \bar{e}_m^o n_m > \bar{e}_c^o n_c.$$

$$\frac{\bar{e}_m^o}{\bar{e}_c^o} = \left[\frac{\tau_m}{\tau_s}(S_1 L_1)^{\phi_m - \phi_c}\right]^{1/2}, \qquad \frac{n_m}{n_c} = \left(\frac{F_m}{F_c}\right)^{\frac{1}{1+\phi_E}}.$$

In the terminal service economy, additional public procedures raise formalization on an unsaturated margin:

$$K_{A,3} > K_{B,3} \quad \Longleftrightarrow \quad S_A^\infty > S_B^\infty \quad \Longleftrightarrow \quad g_A^\infty > g_B^\infty.$$

Common initial states, prices, operating-surplus terms and terminal full-income ratio; symmetric static firms; $\tau_h = 0$, $\Delta = 1$. Nonsaturation requires that an additional public procedure changes formalization for some service firms.

◂ Proposition 3

HSPS: Measurement in Calibration and Cross-Country Data

U.S. calibration: use-based sectoral allocation

- In-house organizational and technical occupations, plus purchased producer services.
- Bachelor's degree or higher defines high skill.
- BEA intermediate-use expenditure shares allocate service suppliers to destination sectors; final sales enter consumer-service value added.

Cross-country accounting: harmonized industry–occupation proxy

$$\frac{\text{ISCO-08 occupations 1–3 in ISIC divisions 62–66, 69–75}}{\text{Employment in market services: ISIC sections G–N, R–S}}$$

ILOSTAT, 2010–2019 averages; 48 countries.

The U.S. allocation assumes common skill composition and labor requirements across an industry's buyers. Aggregate HSPS is an accounting aggregate of sectoral targets.

◂ Calibration ◂ Cross-country results

Calibration: Parameter Estimates and External Benchmarks

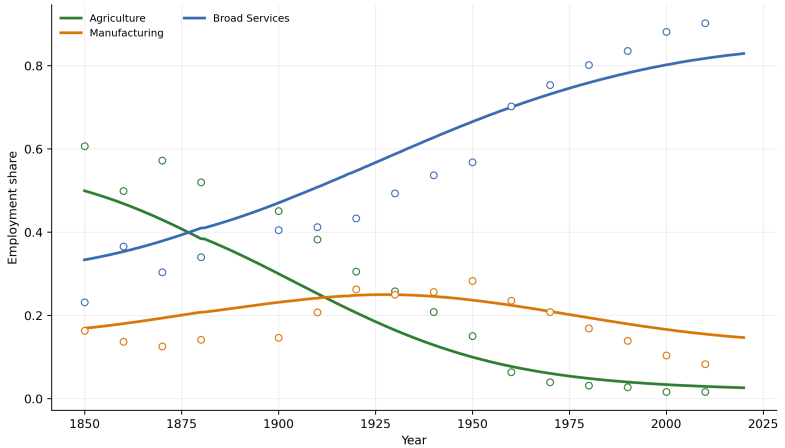
Block	Jointly estimated parameters	Estimates
Fertility	τ, τ_a^n	0.0324, 0.4402
Productivity	$\phi_a, \phi_m, \phi_c, g_{GO}$	0.9984, 0.9192, 0.6905, 0.00318
Skills	$c_{\text{col}}, \alpha_a, \alpha_m, \alpha_c$	0.5682, 0.2434, 0.3300, 0.4100
Demand	ϵ_c, w_a, w_c	1.0105, 0.6510, 3.8233
Organization	R, ξ_a, ξ_c	22.79, 3.5640, 6.8774
Entry	F_a, F_m, F_c	25.872, 187.625, 85.117

Selected assigned parameters	Value / empirical anchor
δ_K	0.09; diffusion benchmark (Mansfield, 1961)
ρ	0.40; maintained adaptation/creation cost ratio
$\bar{N}$	25; recurring coordination uses about 5% of employment
σ_{skill}	0.10; college attainment and earnings-premium changes
η; σ	0.85 (span of control); 0.30 (final-good substitution)

Joint SMM. The nonredundant Jacobian has rank 19; effective rank 17 at a 1% cutoff. Service adaptability is weakly pinned down within its saturation region.

◂ Calibration

Fit to U.S. Structural Change



Lines: model; circles: data. Employment paths are targeted. Manufacturing peak: 24.93% in 1928 (model), 28.28% in 1950 (data).

▸ Targeted fit ◂ Calibration

Calibration: Targeted Scalar Moments

Moment	Data	Model
HSPS share: agriculture	0.3245	0.3248
HSPS share: manufacturing	0.4608	0.4608
HSPS share: consumer services	0.3989	0.3993
Employees per firm: agriculture	7.435	7.441
Employees per firm: manufacturing	49.694	49.717
Employees per firm: services	20.454	20.454
VA productivity growth: agriculture	0.0371	0.0385
VA productivity growth: manufacturing	0.0317	0.0298
VA productivity growth: consumer services	0.0142	0.0131
Aggregate skilled share	0.3299	0.3247
Skilled earnings premium	1.600	1.645
Peak manufacturing-employment share	0.2828	0.2493
Agriculture/service fertility ratio	2.000	1.990

Shares and annual growth rates in fractions. HSPS: 2015–2020; firm size: 2019; productivity growth: 1970–2010. Employment, fertility and GDP paths also enter SMM.

◂ Calibration ◂ U.S. fit

Manufacturing Demand and Equivalent Consumption Wedges

At fixed income and producer prices, define

$$f_m \equiv \frac{w_m^{\mathrm{CF}}}{w_m^\star}, \qquad k_j \equiv \frac{\varepsilon_j}{\varepsilon_m}, \qquad j \in \{a, c\}.$$

Match the relative-demand terms

$$\underbrace{\left(\frac{p_j}{p_m}\right)^{1-\sigma} \frac{w_j}{\left(f_m w_m^\star\right)^{k_j}}}_{\text{Demand shifter}} = \underbrace{\left(\frac{(1+\tau_j)p_j}{p_m}\right)^{1-\sigma} \frac{w_j}{\left(w_m^\star\right)^{k_j}}}_{\text{Consumption wedge}}$$

$$(1+\tau_j)^{1-\sigma} = f_m^{-k_j} \quad \Longleftrightarrow \quad \boxed{1+\tau_j = f_m^{-k_j/(1-\sigma)}}.$$

◂ Experiment ◂ Growth paths

Sensitivity of the Cross-Country Gradients

18 economic variants: all 90 slopes retain their benchmark sign.

	GDP level	GDP growth	Tertiary	HSPS	Population growth
Median absolute change (%)	2.9	8.5	10.2	2.4	4.3
Exact diffusion grid: $\delta_K \in [0.04, 0.14]$					
Minimum change (%)	−13.8	−6.9	−27.8	−22.4	−16.2
Maximum change (%)	0.6	5.0	4.1	1.6	0.6

- Broad variants: adaptation costs, diffusion, scale, substitution, ability dispersion, education costs and knowledge accumulation.
- All signs also survive the wider diffusion grid $\delta_K \in [0.01, 0.15]$.

No recalibration. Broad variants use a common coarser grid. At $\delta_K = 0.02$–0.05, one economy per row has residual discrete-formalization oscillation; precise magnitudes are qualified.

◂ HSPS and growth ◂ Mechanism accounting

When can a temporary intervention change an economy's growth path?

Recalibrate to a median LMIC with small establishments and scarce skills.

Endogenous establishment setup

Before drawing ability a, entrants choose setup intensity h:

$$x = ha, \qquad h \in [\underline{h}, 1].$$

Lower h reduces productivity and the setup inputs needed to enter.

Distortions against productive firms

More productive establishments face larger revenue wedges:

$$1 - \tau_i = \kappa_j x_i^{-d_j}, \qquad d_j = 0.41.$$

Removing d_j restores $h = 1$ and the benchmark entry block.

◂ Policy experiment

Temporary Policies: Costs and Outcomes

Temporary policy	PV cost (% GDP)	CEV (%)	GDPpc (%)	Formalized tasks (pp)	Skilled emp. (pp)	ΔK (level)
Distortions remain						
Skill acquisition scholarships	0.593	2.59	5.56	0.00	0.01	0.000
Formalization support	0.017*	0.18	2.13	0.27	0.13	0.036
Mixed package	0.613	6.95	8.02	0.00	0.02	0.000
HSPS wage subsidy	0.101†	−3.31	−1.51	0.10	0.04	0.009
Manufacturing wage subsidy	0.598	−3.48	−2.14	0.00	−0.01	0.000
Full permanent distortion reform						
Skill acquisition scholarships	0.581	6.92	8.52	1.64	0.29	0.026
Formalization support	0.123*	1.67	8.43	8.63	1.46	0.256
Mixed package	0.600	7.51	10.78	4.48	0.75	0.078
HSPS wage subsidy	0.580	0.57	1.79	3.15	0.55	0.052
Manufacturing wage subsidy	0.597	−0.16	−0.42	0.00	0.00	0.000

PV cost: discounted spending / discounted GDP in 2026–2045. CEV: 2026–2089; other outcomes: 2089, relative to each environment's no-policy path. *100% reimbursement cap. †Fiscal spending jumps from 0.101% to 2.083% across the threshold.

◂ Policy results

Full Reform with the U.S. Entry Block

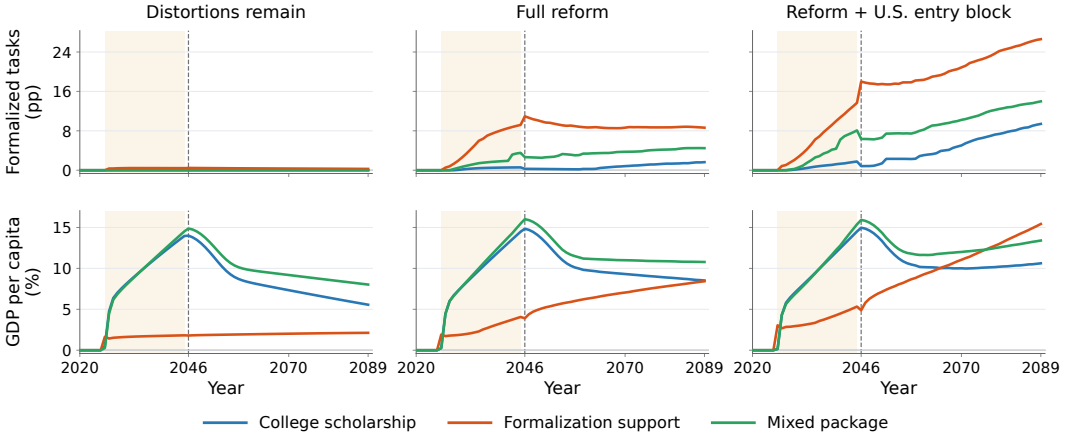
From the common inherited 2026 state: $F_m = 187.625$, $F_c = 85.117$.

Temporary policy	PV cost (% GDP)	CEV (%)	GDPpc (%)	Formalized tasks (pp)	Skilled emp. (pp)	ΔK (level)
Skill acquisition scholarships	0.579	7.17	10.62	9.40	1.41	0.069
Formalization support	0.130*	2.30	15.43	26.60	4.04	0.381
Mixed package	0.585	7.77	13.40	13.98	2.10	0.140
HSPS wage subsidy	0.620	0.70	3.82	11.02	1.63	0.086
Manufacturing wage subsidy	0.597	−0.15	−0.41	0.00	0.00	0.000

Policies operate in 2026–2045. CEV: 2026–2089; other outcomes: 2089. Effects are relative to the no-policy path with full reform and U.S. entry costs. *Formalization support reaches the 100% reimbursement cap.

◂ Policy results

Policy Withdrawal: Formalization and Income



Changes relative to each environment's no-policy path, 2020–2089. Shading: temporary support in 2026–2045; dashed line: unexpected withdrawal in 2046.

Skills and knowledge

Manufacturing History and Subsequent Growth

High peak: 29.88% **Low peak: 11.19%**

		2030	2050	2070
Skilled-employment share (%)	High	34.18	34.64	35.02
	Low	28.74	29.37	29.83
GDP-per-capita growth (% p.a.)	High	2.24	2.22	2.20
	Low	1.99	2.01	2.03
GDP per capita (low peak = 100)	High	139.44	145.57	150.89

Only w_m differs: 3.49 (high), 0.60 (low). Growth is the annualized log change in chain-weighted real GDP per capita over the preceding decade.

◂ Cross-country accounting

Mechanism Accounting

Repeat the manufacturing-demand experiment, changing one mechanism at a time.

	Share of baseline slope (%)				
	HSPS share	Tertiary attainment	GDP per capita	GDPpc growth	Population growth
	2010–2019	2010–2019	2019	2010–2019	1850–2019
Baseline	100.0	100.0	100.0	100.0	100.0
Equal entry barriers, F_j	12.7	−38.4	49.0	40.7	36.0
No knowledge diffusion, $\delta_K = 0$	47.2	26.5	71.3	67.6	67.9
Equal productivity elasticities, ϕ_j	119.1	81.8	76.4	118.7	1.5
Equal adaptability, ξ_j	83.6	77.9	92.7	89.7	93.0
Equal VA skill intensity, α_j	99.7	113.0	108.8	99.8	122.8

Common five-economy support; slopes use each counterfactual's realized manufacturing peak. Other parameters fixed; no recalibration.

▸ Sensitivity ◂ Cross-country accounting

Service Industries in the ETD

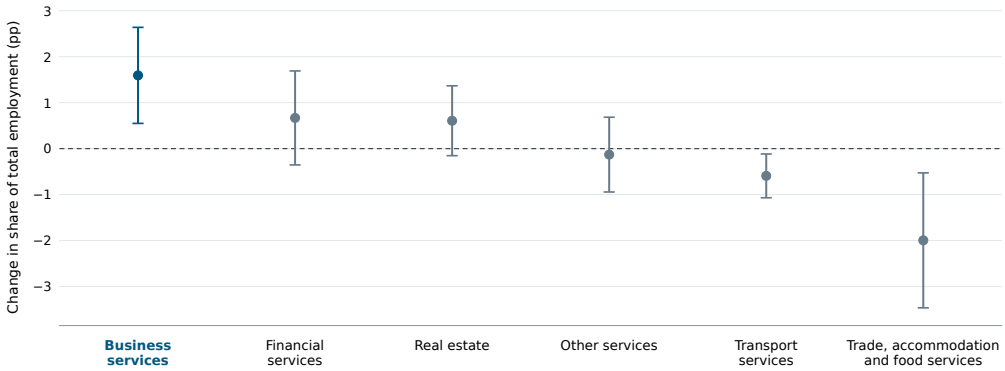
Figure category	ISIC Rev. 4	Activities included: examples
Business services	J, M, N	IT, telecoms and media; accounting, consulting, engineering and R&D; recruitment, equipment rental, cleaning and security.
Financial services	K	Banking, insurance, pension funds and related financial services.
Real estate	L	Property rental, brokerage and management.
Other services	R, S, T, U	Arts, recreation, personal services, household repairs and associations; domestic workers, household activities and extraterritorial bodies.
Transport services	H	Passenger and freight transport, warehousing, postal and courier services.
Trade, accommodation and food services	G, I	Wholesale and retail, motor-vehicle repair, hotels, restaurants and catering.

Industry totals include all occupations. The figure excludes O–Q: public administration, education, health and social work.
Source: GGDC/UNU-WIDER ETD; UN ISIC Rev. 4.

◂ Service-industry decomposition

Stylized Fact 1: Service-Industry Decomposition

Change in employment share, 1990–2018, per 10 pp larger manufacturing peak.



51 economies; separate OLS regressions, no additional controls. Shares of total employment; 95% CIs using HC3 standard errors. Source: ETD and GGDC.

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What Does “High Throughput” Mean?

LOW THROUGHPUT

Textiles & apparel



Scale by adding producers

HIGH THROUGHPUT

Iron & steel



Scale by coordinating a system

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